

Technical Documentation:

Live Quantitative Risk Engine & Portfolio Optimizer

1. Project Overview

The **Quantitative Risk Engine & Portfolio Optimizer** is an automated financial analytics pipeline. It dynamically ingests real-time market data to evaluate risk-adjusted metrics across a multi-asset portfolio. Leveraging Modern Portfolio Theory (MPT), the engine executes a Monte Carlo simulation to plot a Markowitz Efficient Frontier, mathematically identifying the Maximum Sharpe Ratio portfolio allocation.

2. System Architecture & Tech Stack

Component	Technology	Purpose
Data Ingestion	R (yfR)	Extracts live, unpaginated adjusted closing prices via Yahoo Finance APIs.
Data Wrangling	R (dplyr, tidyr)	Cleanses missing values and transforms time-series data into return matrices.
Quantitative Engine	R (Base Math)	Computes covariance matrices, expected returns, and risk scalars.
Visualization (Code)	R (ggplot2)	Renders programmatic scatter plots for exploratory data analysis (EDA).
Business Intelligence	Tableau	Hosts the final executive dashboard for interactive stakeholder presentation.

3. Core Mathematical Framework

The engine relies on institutional-grade financial mathematics to evaluate performance, removing human bias and emotion from the asset allocation process.

- **Daily Logarithmic Returns:** Used for continuous compounding symmetry.

$$r_t = \ln(P_t / P_{t-1})$$

- **Annualized Return:** Scales average daily returns to a standard 252-day trading year.

$$R_{\text{annual}} = (1 + \bar{r})^{252} - 1$$

- **Annualized Volatility (Risk):** Scales the standard deviation of daily returns.

$$\sigma_{\text{annual}} = \sigma_{\text{daily}} \sqrt{252}$$

- **Sharpe Ratio:** Measures risk-adjusted performance (assuming a risk-free rate of zero for simplicity).

$$S = R_{\text{annual}} / \sigma_{\text{annual}}$$

- **Portfolio Variance:** Calculates the aggregate risk of a weighted portfolio, factoring in the covariance between individual assets.

$$\sigma_p^2 = \mathbf{w}^T \Sigma \mathbf{w}$$

(Where $\mathbf{w}$ is the vector of asset weights and Σ is the covariance matrix of asset returns).

4. Pipeline Execution: Phase by Phase

Phase 1: API Ingestion & Risk Profiling

The script initializes by calling the Yahoo Finance API for five predefined tickers (AAPL, MSFT, NVDA, AMZN, GOOGL) from January 1, 2023, to the current system date.

1. **Transformation:** Converts raw adjusted closing prices into daily log returns using `dplyr::lag()`.
2. **Risk Profiling:** Groups the data by ticker to aggregate average returns, standard deviation, and the 95% Historical Value-at-Risk (VaR).
3. **Output:** A localized R dataframe (`portfolio_risk_summary`) ranking assets by pure Sharpe Ratio efficiency.

Phase 2: Tableau BI Export

To bridge the gap between statistical modeling and business intelligence, the pipeline formats and exports two optimized CSV files:

- **tableau_stock_time_series.csv:** Contains the cumulative product of daily returns representing a base \$1.00 investment growth over time.
- **tableau_portfolio_risk_summary.csv:** Contains the annualized metrics for X/Y scatter plotting.

Phase 3: The Monte Carlo Optimization Engine

The engine converts the time-series return data into an $N \times M$ matrix to compute the asset Covariance Matrix (Σ).

1. **Simulation:** Iterates 10,000 times. For each loop, it generates a random vector of asset weights summing to 1.0.
2. **Calculation:** Applies the randomized weights to the mean return vector and covariance matrix using linear algebra to find the simulated portfolio's aggregate Return, Volatility, and Sharpe Ratio.
3. **Isolation:** Filters the 10,000-row output table to isolate the single row containing the `max(Sharpe_Ratio)`.

5. Final Optimization Results & Business Intelligence

The simulation empirically rejected an equal-weight distribution strategy. To achieve the absolute highest expected return per unit of volatility taken, the algorithm designated the following optimal weights:

Ticker	Company	Target Allocation	Strategic Role
NVDA	Nvidia	65.8%	Primary Growth Driver
GOOGL	Alphabet	24.1%	Core Volatility Anchor
AMZN	Amazon	6.9%	Supplementary Growth
MSFT	Microsoft	1.9%	Marginal Allocation
AAPL	Apple	1.3%	Marginal Allocation

Simulated Portfolio Projections:

- Expected Annual Return: **86.7%**
- Annualized Volatility (Risk): **36.6%**
- Maximized Sharpe Ratio: **2.37**

6. Setup & Execution Instructions

To replicate this environment locally:

1. Ensure R and RStudio are installed.
2. Install required dependencies via the console: `install.packages(c("yfR", "tidyverse", "ggplot2"))`
3. Execute the master R script. The yfR package will automatically fetch real-time data, replacing the static historical data in the calculations.
4. Locate the exported `.csv` files in the working directory to refresh the Tableau dashboard.